

DIHWorld Audio

Dynamics / Guard Technical Manual

Current product versions, controls, presets, meters, and placement

DIHWorld Audio builds connected production systems around the Imhotep 369 Engine—a Triadic Harmonic Lattice—and Ma'at 357 / MomentumBend. Phi-derived control and the canonical Phi/2 ceiling preserve movement, balance, and musical intent without brickwall limiting.

This edition documents 4 current products. Release numbers are taken from the installed DIHWorld Audio Audio Units verified on 2026-10-05; they are product versions, not a shared catalog version.

Canon reference: normalized $\Phi/2 = 0.809016994$. MomentumBend shapes motion and containment; the ceiling is a canonical boundary, not permission for hard limiting.

Operating law: begin from reset or a purpose-matched preset, compare at equal loudness, change one system at a time, and judge the result in the full arrangement.

ANUBIS

RELEASE	FAMILY	PLACEMENT
v1.0.1	Dynamics / Guard	Insert Safe

Dynamics gate and guard with threshold, ratio, attack, hold, release, range, a named stereo host Sidechain bus, independent Sidechain Key and Sidechain Duck operation, protected I/O, and parallel mix.

Recommended use: guard/dynamics

Customer formats: AU, VST3, Standalone

Exposed parameter surface

Parameter ID	Display name	Type
input	Input	Float
mix	Mix	Float
output	Output	Float
power	Power	Bool
threshold	Threshold	Float
ratio	Ratio	Float
attack	Attack	Float
hold	Hold	Float
release	Release	Float
range	Range	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
extkey	Sidechain Key	Bool
sduck	Sidechain Duck	Bool
gainmatch	Gain Match	Bool

Factory starting points — source-derived purpose

TRANSPARENT PASS [Stereo]

TRANSPARENT PASS — Returns the gate to a transparent pass so Threshold and envelope behavior can be judged from the source. Stereo variant: keep the linked center intact and audition width in the full mix.

VOCAL BREATH GUARD [Stereo]

VOCAL BREATH GUARD — Keeps vocal breaths and consonants natural while the guard controls spill and room energy. Stereo variant: keep the linked center intact and audition width in the full mix.

KICK KEY GATE [Stereo]

KICK KEY GATE — Uses the key/duck detector to make kick or sidechain energy the envelope decision. Stereo variant: keep the linked center intact and audition width in the full mix.

TEHUTI DUCK GUARD [Stereo]

TEHUTI DUCK GUARD — Uses the key/duck detector to make kick or sidechain energy the envelope decision. Stereo variant: keep the linked center intact and audition width in the full mix.

KEY + DUCK LOCK [Stereo]

KEY + DUCK LOCK — Uses the key/duck detector to make kick or sidechain energy the envelope decision. Stereo variant: keep the linked center intact and audition width in the full mix.

DRUM TIGHT GUARD [Stereo]

DRUM TIGHT GUARD — Tightens drum spill with Attack/Hold/Release chosen to keep the hit intact. Stereo variant: keep the linked center intact and audition width in the full mix.

MONO WALL HOLD [Mono]

MONO WALL HOLD — Applies the guard to the selected channel layout while preserving the channel-mode decision. Mono variant: collapse the field and

use it as a center-focused insert.

DUAL MONO LOCK [Dual Mono]

DUAL MONO LOCK — Applies the guard to the selected channel layout while preserving the channel-mode decision. Dual-mono variant: preserve independent left/right decisions for asymmetric sources.

Meters and visualizers

IN: Main-input RMS energy. Shows the program feeding the gate.

GATE: Current gate passage gain. OPEN is full passage; lower values mean attenuation.

KEY: External sidechain RMS energy. Moves only when a host sidechain is routed.

DUCK: Sidechain-driven gain reduction. More fill means deeper ducking of the main signal.

OUT: Post-gate RMS energy. Compare with IN to judge actual attenuation.

Transfer graph: Threshold, envelope, and Range floor. Shows where the signal opens and how far closure may fall.

Correlation waves: Left/right phase relationship. Gold is coherent, purple is spread, red warns of anti-phase risk.

OTIS

RELEASE	FAMILY	PLACEMENT
v1.0.1	Dynamics / Guard	Insert Safe

3-6-9 macro dynamics processor: Micro Peak controls short events, Meso Density shapes body, and Macro Level governs the long envelope.

Recommended use: dynamics support; secondary silver mastering-support dot

Customer formats: AU, VST3, Standalone

Exposed parameter surface

Parameter ID	Display name	Type
trumpet_l	Micro Peak	Float
trumpet_c	Meso Density	Float
trumpet_r	Macro Level	Float
makeup_gain	Makeup Gain	Float
gainmatch	Gain Match	Bool

Factory starting points — source-derived purpose

TRANSPARENT PASS [Stereo]

TRANSPARENT PASS — Returns the three timescales to a transparent pass for judging the source envelope. Stereo variant: keep the linked center intact and audition width in the full mix.

VOCAL LEVELER [Stereo]

VOCAL LEVELER — Lets Micro Peak catch consonant spikes while Meso Density and Macro Level keep the phrase even. Stereo variant: keep the linked center intact and audition width in the full mix.

DIALOG STEADY [Stereo]

DIALOG STEADY — Lets Micro Peak catch consonant spikes while Meso Density and Macro Level keep the phrase even. Stereo variant: keep the linked center intact and audition width in the full mix.

DRUM BUS CATCH [Stereo]

DRUM BUS CATCH — Catches drum peaks without erasing the hit; use Meso Density only as the bus settles. Stereo variant: keep the linked center intact and audition width in the full mix.

MIX BUS GLUE [Stereo]

MIX BUS GLUE — Applies gentle multi-timescale glue to a bus; keep Gain Match independent of this preset. Stereo variant: keep the linked center intact and audition width in the full mix.

TRANSIENT GUARD [Stereo]

TRANSIENT GUARD — Favors Micro Peak control and leaves the phrase-level Macro lane more open. Stereo variant: keep the linked center intact and audition width in the full mix.

MONO LEVEL HOLD [Mono]

MONO LEVEL HOLD — Uses the OTIS leveler in the selected channel layout; compare both channels at matched output. Mono variant: collapse the field and use it as a center-focused insert.

DUAL MONO CONTROL [Dual]

DUAL MONO CONTROL — Uses the OTIS leveler in the selected channel layout; compare both channels at matched output. Dual variant: keep channel decisions independent when the source is asymmetric.

Meters and visualizers

INPUT: Input peak in dB. Confirms level feeding all three time scales.

RESPONSE: Strongest active 3/6/9 reduction. Shows the greatest current reduction across Micro, Meso, or Macro.

OUTPUT: Output peak in dB. Use with INPUT and Gain Match for fair comparison.

3 / Micro activity: Fast transient reduction. Responds to short peaks and front edges.

6 / Meso activity: Body and phrase-density reduction. Responds to medium-duration energy.

9 / Macro activity: Long-envelope reduction. Responds to sustained level movement.

DeBo

RELEASE	FAMILY	PLACEMENT
v3.6.9	Dynamics / Guard	Insert Safe

Pressure-activated dynamics processor with intentional aggressive Drive excitation, pressure-dependent density, controlled Maximize gain, and ceiling containment. DEBO Output controls delivery after the drive/pressure stage, so deliberate distortion can remain while output is reduced. In the DEBO -> DEPTH_BOOST_PRO hybrid, DEBO creates and controls character first; Depth Boost Pro then places it through DIH Field State and can apply optional post-field Spatial Gain reduction.

Recommended use: pressure-activated dynamics, aggressive drive, density, and controlled output

Customer formats: AU, VST3, Standalone

Exposed parameter surface

Parameter ID	Display name	Type
POWER	Power	Bool
FLOWER	Flower of Life	Bool
GAINMATCH	Gain Match	Bool
INPUT	Input	Factory
DRIVE	Drive	Factory
MAXIMIZE	Maximize	Factory
DEPTH	Depth	Factory
CLARITY	Clarity	Factory
OUTPUT	Output	Factory

Factory starting points — source-derived purpose

NO BRICKS [Stereo]

NO BRICKS — Open DEBO density with zero maximize gain: add Drive/Depth/Clarity while explicitly avoiding a brick-wall loudness target. Factory recipe: Drive 56%, Maximize Db 0 dB, Depth 52%, Clarity 58%. Stereo variant: keep the linked center intact and audition width in the full mix.

VOCAL HOLD [Stereo]

VOCAL HOLD — A vocal containment recipe with moderate Drive/Depth and only a small maximize lift, keeping the phrase stable without flattening it. Factory recipe: Drive 58%, Maximize Db 0.8 dB, Depth 60%, Clarity 46%. Stereo variant: keep the linked center intact and audition width in the full mix.

BRICK TAP [Stereo]

BRICK TAP — A measured first touch of maximize pressure for a source that needs forwardness but still has room to breathe. Factory recipe: Drive 58%, Maximize Db 0.5 dB, Depth 54%, Clarity 50%. Stereo variant: keep the linked center intact and audition width in the full mix.

WALL PRESSURE [Stereo]

WALL PRESSURE — Raises pressure and depth for a denser bus; use it when the mix needs a firmer wall, then check the vocal and kick. Factory recipe: Drive 60%, Maximize Db 0.8 dB, Depth 58%, Clarity 50%. Stereo variant: keep the linked center intact and audition width in the full mix.

SHOULDER CHECK [Stereo]

SHOULDER CHECK — A mid-strength density check that tests whether the arrangement can carry more edge without losing clarity. Factory recipe: Drive 62%, Maximize Db 1.1 dB, Depth 62%, Clarity 50%. Stereo variant: keep the linked center intact and audition width in the full mix.

NO SCISSORS [Stereo]

NO SCISSORS — Higher Drive/Depth with moderate maximize keeps peaks from feeling sharply clipped while the body becomes denser. Factory recipe: Drive 64%, Maximize Db 1.4 dB, Depth 66%, Clarity 48%. Stereo variant: keep the linked center intact and audition width in the full mix.

BREAKTHROUGH [Stereo]

BREAKTHROUGH — A strong forward push for a chorus or impact section; reduce Maximize if the transients begin to sound cut off. Factory recipe: Drive 66%, Maximize Db 1.8 dB, Depth 70%, Clarity 46%. Stereo variant: keep the linked center intact and audition width in the full mix.

RED LINE WATCH [Stereo]

RED LINE WATCH — The most assertive DEBO recipe: use it as a translation stress test or special-effect push, not as the default master. Factory recipe: Drive 68%, Maximize Db 2.2 dB, Depth 74%, Clarity 44%. Stereo variant: keep the linked center intact and audition width in the full mix.

PHI GUARD [Stereo]

PHI GUARD — Depth is set to the canonical Phi-derived guard point while maximize stays at zero; use it for controlled density without loudness chasing.

Factory recipe: Drive 62%, Maximize Db 0 dB, Depth 81%, Clarity 48%. Stereo variant: keep the linked center intact and audition width in the full mix.

VOCAL ANCHOR [Stereo]

VOCAL ANCHOR — Use it to add pressure and forward motion without a brick-wall default: the recipes move from open containment to deliberate loudness support, with Clarity preserving the important edge. Use this DEBO program as a controlled starting recipe for “VOCAL ANCHOR”; adjust the exposed processor controls in small, level-matched moves and keep the source role named by the preset in view. Factory recipe: Drive 59%, Maximize Db 0.6 dB, Depth 62%, Clarity 42%. Stereo variant: keep the linked center intact and audition width in the full mix.

DRUM REINFORCE [Stereo]

DRUM REINFORCE — Use it to add pressure and forward motion without a brick-wall default: the recipes move from open containment to deliberate loudness support, with Clarity preserving the important edge. Use this DEBO program as a controlled starting recipe for “DRUM REINFORCE”; adjust the exposed processor controls in small, level-matched moves and keep the source role named by the preset in view. Factory recipe: Drive 64%, Maximize Db 1 dB, Depth 55%, Clarity 30%. Stereo variant: keep the linked center intact and audition width in the full mix.

MIX BUS SETTLE [Stereo]

MIX BUS SETTLE — Use it to add pressure and forward motion without a brick-wall default: the recipes move from open containment to deliberate loudness support, with Clarity preserving the important edge. Use this DEBO program as a controlled starting recipe for “MIX BUS SETTLE”; adjust the exposed processor controls in small, level-matched moves and keep the source role named by the preset in view. Factory recipe: Drive 60%, Maximize Db 0.4 dB, Depth 52%, Clarity 38%. Stereo variant: keep the linked center intact and audition width in the full mix.

PARALLEL BREACH [Stereo]

PARALLEL BREACH — Use it to add pressure and forward motion without a brick-wall default: the recipes move from open containment to deliberate loudness support, with Clarity preserving the important edge. Use this DEBO program as a controlled starting recipe for “PARALLEL BREACH”; adjust the exposed processor controls in small, level-matched moves and keep the source role named by the preset in view. Factory recipe: Drive 68%, Maximize Db 2 dB, Depth 70%, Clarity 40%. Stereo variant: keep the linked center intact and audition width in the full mix.

MASTER FIELD [Stereo]

MASTER FIELD — Use it to add pressure and forward motion without a brick-wall default: the recipes move from open containment to deliberate loudness support, with Clarity preserving the important edge. Use this DEBO program as a controlled starting recipe for “MASTER FIELD”; adjust the exposed processor controls in small, level-matched moves and keep the source role named by the preset in view. Factory recipe: Drive 62%, Maximize Db 0.5 dB, Depth 58%, Clarity 34%. Stereo variant: keep the linked center intact and audition width in the full mix.

Meters and visualizers

RMS: Sustained signal body. Shows average energy driving the pressure model.

GUARD PRESSURE: Combined density/containment demand. Higher values mean DEBO is working harder.

PEAK: Fast peak relative to ceiling. Approaching the top warns of limited remaining space.

GOVERNOR: Gain held back by protection. More activity means stronger safety containment.

Pressure window / face: RMS-driven pressure stage. The visual advances with real pressure; red is the ceiling warning state.

LookOut

RELEASE	FAMILY	PLACEMENT
v1.0.1	Dynamics / Guard	Insert Safe

Dual-envelope transient/sustain shaper. Attack and Sustain directly move the level of the transient and sustain portions of the signal, up to +/-18 dB each, ahead of the DIH tone-color stage. Its Eye of Heru state also gives live advisory guidance on transient/sustain balance while it shapes.

Recommended use: dual-envelope transient/sustain shaper; Attack/Sustain directly move transient and sustain gain up to +/-18dB; Eye of Heru advisory guidance is secondary, not the primary function; cyan primary dot with green observation and silver mastering-support dots

Customer formats: AU, VST3, Standalone

Exposed parameter surface

Parameter ID	Display name	Type
power	Power	Factory
attack	Attack	Factory
sustain	Sustain	Factory
gainmatch	Gain Match	Factory

Factory starting points — source-derived purpose

- NEUTRAL SENTINEL [Stereo]**
NEUTRAL SENTINEL — Attack and Sustain at unity: an observation reset for hearing the source before choosing peak or tail emphasis. Factory recipe: Attack 0%, Sustain 0%. Stereo variant: keep the linked center intact and audition width in the full mix.
- VOCAL FRONT GUARD**
VOCAL FRONT GUARD — Moderate Attack with a shorter Sustain watches vocal fronts while leaving phrase movement intact. Factory recipe: Attack 46, Sustain 18.
- DRUM SNAP LOCK**
DRUM SNAP LOCK — Fast Attack and negative Sustain protect the drum transient and stop the tail from smearing the next hit. Factory recipe: Attack 88, Sustain -64.
- KICK BODY ANCHOR**
KICK BODY ANCHOR — Negative Attack with strong positive Sustain favors the kick body and low envelope instead of the initial click. Factory recipe: Attack -62, Sustain 90.
- PERCUSSION EDGE**
PERCUSSION EDGE — High Attack and negative Sustain isolate percussion edges for crisp placement without a broad level lift. Factory recipe: Attack 74, Sustain -86.
- ROOM TAIL WATCH**
ROOM TAIL WATCH — Negative Attack with positive Sustain keeps an ear on room decay and tail build-up rather than only the first hit. Factory recipe: Attack -78, Sustain 72.
- MIX BUS COMPOSURE**
MIX BUS COMPOSURE — Moderate positive Attack/Sustain observes the bus as a whole and keeps movement calm without hard clamping. Factory recipe: Attack 28, Sustain 48.
- FINAL PEAK SENTINEL**
FINAL PEAK SENTINEL — High Attack with lightly reduced Sustain is a final peak watch; use it to find events, not to replace a limiter. Factory recipe: Attack 62, Sustain -18.

Meters and visualizers

- Transient:** Detected front-edge energy. Use to judge whether Attack is emphasizing or restraining hits.
- Sustain:** Detected body and tail energy. Use to judge Sustain changes to ring, room, and body.
- Alert:** Potential transient/sustain imbalance. Higher values suggest the balance needs attention.
- Miss:** Underrepresented event energy. Indicates material that may be disappearing.

Stress: Over-concentrated envelope pressure. Reduce shaping intensity when stress remains high.

Composure: Stable transient/sustain balance. Higher is calmer and more controlled.

Eye of Heru: Combined advisory state. It reports balance; it does not move controls automatically.